Syntax of defining a dictionary:

```
Dictionary_Name = {     Key_1: Value_1,
                        Key_2:Value_2,
                        ……..
                        Key_n:Value_n
                        }
```

Key in the dictionary must be unique case sensitive and can be of any valid Python type.

9.4.1 Creating a Dictionary

```
# Empty dictionary
    Dict1 = { }

# Dictionary with Key
    Dict_Stud = { 'RollNo': '1234', 'Name':'Murali', 'Class':'XII', 'Marks':'451'}
```

9.4.2 Dictionary Comprehensions

In Python, comprehension is another way of creating dictionary. The following is the syntax of creating such dictionary.

Syntax

```
Dict = { expression for variable in sequence [if condition] }
```

The if condition is optional and if specified, only those values in the sequence are evaluated using the expression which satisfy the condition.

Example

```
Dict = { x : 2 * x for x in range(1,10)}
```

Output of the above code is

```
{1: 2, 2: 4, 3: 6, 4: 8, 5: 10, 6: 12, 7: 14, 8: 16, 9: 18}
```

9.4.3 Accessing, Adding, Modifying and Deleting elements from a Dictionary

Accessing all elements from a dictionary is very similar as Lists and Tuples. Simple print function is used to access all the elements. If you want to access a particular element, square brackets can be used along with key.